# Basin Shaping for Opponent-Aware Learning: Local Meta-MAPG Convergence with Restarts

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## Abstract

Policy-gradient methods converge locally near stable Nash policies in general stochastic games, but the standard theory does not anticipate opponent learning. LOLA and Meta-MAPG add gradients through other agents' future updates and change equilibrium selection in practice. We give a two-phase stochastic-approximation account of this mechanism. Phase 1 uses a constant Meta-MAPG correction as a basin-entry device; if the shaped field improves local drift by $\lambda\mu_M$, the certified entry time shrinks by the same factor. Phase 2 anneals shaping to zero, so Giannou et al.'s local theorem applies to stable Nash equilibria of the original game. Independent restarts globalise the result with an almost-sure finite-restart guarantee and a geometric tail bound. The experiments match the theorem: peer-learning changes equilibrium selection, restart budget selects equilibria, basin maps expand under opponent-aware shaping, and a small MLP IPD check shows the effect is not purely tabular.

## 1. Introduction

Learning in stochastic games is hard because every learner changes the environment faced by the others. Giannou et al. (2022) prove local convergence of policy gradients to stable Nash policies under strategic stability and second-order sufficiency. This is the right baseline. It still treats opponent learning as background noise, not as a quantity to shape.

Opponent-aware policy gradients put this dependence into the update. LOLA differentiates through an opponent's anticipated step (Foerster et al., 2018a; Letcher et al., 2019). Meta-MAPG gives three terms: the current-policy gradient, an own-learning gradient, and a peer-learning gradient for the effect of current behaviour on other agents' future policies (Kim et al., 2021). These terms matter empirically. Their stochastic-approximation status is less clear.

We prove three claims. First, under bounded rewards, smooth policies, controlled unroll length, and standard sampling assumptions, the Meta-MAPG update is a stochastic approximation to the base game-gradient field plus controlled opponent-aware terms. Second, constant shaping and convergence need not conflict: use constant Meta-MAPG shaping only for finite-time basin entry, then anneal it so the limiting stationary points remain the stable Nash equilibria of the original game.

Third, independent restarts globalise the local theorem. Local convergence depends on basin mass; normal-form games can have cycles and several attractors. With a full-support restart distribution on a compact parameter set, each epoch has a fixed positive probability of landing in the basin. This gives almost-sure finite restart success and an expected restart bound. Restart budget becomes equilibrium-selection budget. Experiments in Stag Hunt and iterated Prisoner's Dilemma test the theorem's ingredients: component ablations, restart-budget selection, basin volume, and estimator variance.

**Positioning.** Prior work builds opponent-shaping algorithms. LOLA differentiates through an opponent update (Foerster et al., 2018a); SOS stabilises shaping in differentiable games (Letcher et al., 2019); COLA enforces consistency of mutual shaping (Willi et al., 2022); POLA adds a proximal view (Zhao et al., 2022); M-FOS learns long-horizon shaping policies without a differentiable opponent model (Lu et al., 2022). We ask a narrower question: when does finite-unroll Meta-MAPG fit stochastic approximation, improve basin entry, and preserve the asymptotic Nash target?

## 2. Setup

### 2.1. Stochastic Games

We consider finite discounted stochastic games

$$\mathcal{G} = (N, \mathcal{S}, \{\mathcal{A}_i\}_{i\in N}, P, \{r_i\}_{i\in N}, \rho, \gamma_{\text{disc}}),$$

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*Table 1.* Positioning among opponent-aware methods. "Nash-preserving" means the stated theory preserves the original game's stable Nash target rather than merely optimising a shaped meta-objective.

| Method | Convergence guarantee | Shaping mechanism | Opponent model | Nash-preserving |
|---|---|---|---|---|
| LOLA | No general | 1-step gradient | White-box | No |
| SOS | Differentiable games | Stable OS | White-box | SFP-focused |
| COLA | Simple-game stability | Consistent OS | White-box | No |
| POLA | Empirical/proximal | Proximal OS | Learned | Not primary |
| M-FOS | Empirical | Meta-policy OS | No | No |
| This work | Local SA + restarts | Two-phase Meta-MAPG | Sample estimates | Yes, by annealing |

with finite state space $\mathcal{S}$, finite action sets $\mathcal{A}_i$, transition kernel $P$, bounded rewards $|r_i| \le R_{\max}$, initial-state distribution $\rho$, and discount factor $\gamma_{\text{disc}} \in [0, 1)$. Each agent has a differentiable stationary Markov policy $\pi_{\phi_i}(a_i \mid s)$ with parameter $\phi_i \in \mathbb{R}^{d_i}$; the joint parameter is $\phi = (\phi_i)_{i=1}^N \in \mathbb{R}^d$. The value of agent $i$ is

$$J_i(\phi) = \mathbb{E}_{\rho,\pi_\phi}\left[\sum_{t\ge 0} \gamma_{\text{disc}}^t r_i(s_t, a_t)\right],$$

and the standard game-gradient field is

$$v(\phi) = \left(\nabla_{\phi_i} J_i(\phi)\right)_{i=1}^N.$$

First-order stationary points of this field correspond to Nash policies under the gradient-dominance property used by Giannou et al. (2022); finite stochastic games possess stationary Nash equilibria (Fink, 1964).

**Definition 2.1** (Stable Nash and SOS)**.** A Nash parameter $\phi^*$ is stable if

$$\langle v(\phi), \phi - \phi^* \rangle < 0$$

for all $\phi \ne \phi^*$ sufficiently close to $\phi^*$. It is second-order stationary (SOS) if there exist $\mu > 0$ and $\rho > 0$ such that

$$\langle v(\phi), \phi - \phi^* \rangle \le -\mu \|\phi - \phi^*\|^2$$

whenever $\|\phi - \phi^*\| \le \rho$.

The SOS condition is the quantitative stability condition that yields an attracting neighbourhood and a rate in Giannou et al. (2022). We use it as the local anchor.

### 2.2. Meta-MAPG Field

Meta-MAPG views learning as a Markov chain of joint policy parameters $\phi_0, \dots, \phi_L$, where each inner transition is produced by a policy-gradient update. For agent $i$, Kim et al. (2021) decompose the meta-gradient into

$$G_i^{\text{MAPG}} = G_i^{\text{cur}} + G_i^{\text{own}} + G_i^{\text{peer}}.$$

The current term is the usual policy-gradient score term. The own-learning term differentiates through how $\phi_i^0$ affects agent $i$'s adapted parameters $\phi_i^{1:L}$. The peer-learning term differentiates through how $\phi_i^0$ affects other agents' future parameters $\phi_{-i}^{1:L}$. The peer term is the Meta-MAPG analogue of LOLA-style opponent shaping; the own term is absent from pure LOLA.

We write the population opponent-aware field as

$$F_L(\phi) = v(\phi) + M_L^{\text{own}}(\phi) + M_L^{\text{peer}}(\phi),$$

where $L$ is the unroll length. In the convergence theorem we use the annealed update

$$\phi_{n+1} = \text{Proj}_{\mathcal{K}}\Big[\phi_n + \alpha_n \left(\hat{v}_n + \lambda_n \hat{M}_{L_n,n}\right)\Big], \quad (1)$$

Here $\mathcal{K} \subset \mathbb{R}^d$ is compact, $\hat{M}_{L_n,n}$ estimates the combined own/peer correction, and $\lambda_n$ controls the asymptotic shaping strength. Projection can be replaced by any standard boundedness mechanism. We state it explicitly so the projection lemma of Kushner & Clark (1978) and the ODE tracking argument of Borkar (2008) apply without extra tightness work.

**Definition 2.2** (Nash-consistent shaping)**.** Let $U$ be a neighbourhood of a stable Nash set $\mathcal{N}$ of the base game-gradient field $v$. A scheduled opponent-aware correction $\lambda_n M_{L_n}$ is Nash-consistent on $U$ if its contribution to the stochastic-approximation recursion is either summable,

$$\sup_{\phi\in U} \|\lambda_n M_{L_n}(\phi)\| \le \beta_n, \qquad \sum_n \alpha_n \beta_n < \infty,$$

so the limiting ODE on $U$ is $\dot{\phi} = v(\phi)$, or else its non-vanishing limit field $v + M$ has exactly the same isolated SOS equilibria in $U$ as $v$ with the same local attracting neighbourhoods. The convergence theorem uses the first, annealed case; the second case only records when constant shaping is locally equilibrium-preserving.

## 3. Assumptions

The assumptions separate standard stochastic approximation from the Meta-MAPG estimator step.

**Assumption 1** (Step sizes and bounded iterates)**.** The step sizes satisfy $\alpha_n > 0$, $\sum_n \alpha_n = \infty$, and $\sum_n \alpha_n^2 < \infty$.

Iterates remain in a compact set $\mathcal{K}$, either by projection in (1) or by an equivalent stability mechanism.

**Assumption 2** (Estimator decomposition)**.** There is a locally Lipschitz effective field $F$—equal to $v$ in the annealed case and to an equilibrium-preserving limit $v + M$ in the non-vanishing case (Theorem 2.2)—such that the sampled update $g_n = \hat{v}_n + \lambda_n \hat{M}_{L_n,n}$ admits the decomposition

$$g_n = F(\phi_n) + \xi_{n+1} + b_n,$$

where $\{\xi_{n+1}\}$ is a martingale-difference sequence adapted to $\mathcal{F}_n$ with

$$\mathbb{E}[\xi_{n+1} \mid \mathcal{F}_n] = 0, \qquad \mathbb{E}[\|\xi_{n+1}\|^2 \mid \mathcal{F}_n] \le \sigma^2,$$

and $\|b_n\| \le C\beta_n$ for a deterministic sequence $\beta_n \ge 0$.

**Assumption 3** (Truncation and modeling bias)**.** The unroll schedule $L_n$ and opponent-modeling procedure produce a residual finite-unroll/modeling error that is absorbed into the bias sequence $\beta_n$ of Assumption 2. The combined bias satisfies $\beta_n \to 0$.

**Assumption 4** (Summability and annealing)**.** Step sizes, bias, and shaping coefficients jointly satisfy

$$\sum_n \alpha_n \beta_n < \infty, \qquad \sum_n \alpha_n \lambda_n < \infty,$$

with $M_{L_n}$ uniformly bounded on $\mathcal{K}$. Under these conditions the limiting ODE of (1) is the base game-gradient flow $\dot{\phi} = v(\phi)$, so the opponent-aware correction is asymptotically transient. Phase 1 of Theorem 4.4 relaxes the $\sum_n \alpha_n \lambda_n < \infty$ requirement to a constant $\lambda_n = \lambda_c$ over a finite-time window only; phase 2 restores it.

**Assumption 5** (Strategic stability)**.** The stable set $\mathcal{N}$ consists of isolated SOS Nash equilibria satisfying Theorem 2.1. In the non-annealed local geometry statement, the shaped field $F = v + \lambda M$ satisfies the analogous SOS condition around the same equilibrium.

*Remark* 3.1. The convergence target is the original game's stable Nash set $\mathcal{N}$. If shaping is kept constant, meta-objectives can have non-Nash fixed points. The theorem is therefore annealed, or explicitly equilibrium-preserving. Constant shaping is used for finite-time basin geometry in Theorem 4.3, not to claim that arbitrary meta-game stationary points are Nash.

## 4. Main Result

**Lemma 4.1** (Estimator decomposition and noise control)**.** *Under bounded rewards, twice differentiable policies with bounded score and Hessian on $\mathcal{K}$, and finite second moments of the likelihood-ratio estimators, the finite-unroll Meta-MAPG estimator satisfies Assumption 2. In particular, the current-policy, own-learning, and peer-learning components can each be written as a conditional mean plus a martingale-difference error; finite unroll and opponent-modeling truncation contribute the bias $b_n$.*

**Lemma 4.2** (Local Meta-MAPG convergence)**.** *Suppose Assumptions 1–5 hold and no restarts occur. If $\phi_0$ lies in the attracting neighbourhood of an SOS Nash equilibrium $\phi^* \in \mathcal{N}$, then*

$$\phi_n \to \phi^* \qquad \textit{almost surely.}$$

*Moreover, under the quantitative bias and variance exponents of Giannou et al. (2022), the usual local squared-distance rate is inherited.*

**Proposition 4.3** (Local drift improvement under shaping)**.** *Fix an SOS Nash equilibrium $\phi^*$ of the base game-gradient field $v$ with local drift constant $\mu > 0$. Let $M$ be the population Meta-MAPG correction, assume both $v$ and $M$ are $C^1$ near $\phi^*$ with $M(\phi^*) = 0$, and write $J_v = Dv(\phi^*)$, $J_M = DM(\phi^*)$. Define*

$$\mu_M := -\lambda_{\max}\left(\tfrac{J_M + J_M^\top}{2}\right).$$

*Then for every $\lambda \ge 0$, the shaped field $F_\lambda = v + \lambda M$ satisfies the linearized drift bound*

$$-\lambda_{\max}\left(\tfrac{J_\lambda + J_\lambda^\top}{2}\right) \ge \mu + \lambda\mu_M, \qquad J_\lambda := J_v + \lambda J_M, \quad (2)$$

*by Weyl's inequality on Hermitian matrices. If in addition $DF_\lambda$ is $L_\lambda$-Lipschitz on $\mathcal{B}_\rho(\phi^*)$, then on the ball of radius $\rho_\lambda := (\mu + \lambda\mu_M)/(2L_\lambda)$,*

$$\langle F_\lambda(\phi), \phi - \phi^* \rangle \le -\tfrac{1}{2}(\mu + \lambda\mu_M)\|\phi - \phi^*\|^2,$$

*so the certified SOS radius around $\phi^*$ scales at least linearly in $\lambda$ whenever $\mu_M > 0$. The sign of $\mu_M$ is game-dependent; if $\mu_M \le 0$, opponent-aware shaping does not enlarge the certified local basin.*

**Theorem 4.4** (Two-phase Meta-MAPG)**.** *Fix an SOS Nash equilibrium $\phi^*$ and let $\mathcal{B}_\rho = \{\phi : \|\phi - \phi^*\| \le \rho\}$ be a local region on which the shaped field $F_{\lambda_c} = v + \lambda_c M$ satisfies the SOS drift condition with constant $\mu_{\lambda_c} > 0$. Run Meta-MAPG in two phases. In phase 1, use constant shaping $\lambda_n = \lambda_c$ and constant step size $\alpha_0$ for $N_0$ iterations; if $\phi_{N_0} \notin \mathcal{B}_{\rho/2}$, declare the epoch failed. In phase 2, started only from $\mathcal{B}_{\rho/2}$, use $\alpha_n = \alpha_0 n^{-p}$ and $\lambda_n = \lambda_c n^{-q}$ with $1/2 < p \le 1$ and $q > 1 - p$, together with Assumption 3.*

*If the phase-1 estimator has conditional variance at most $\sigma^2$ and bias norm at most $\beta_0$, then there is a constant $C$ depending on local bounds of $F_{\lambda_c}$ such that*

$$\Pr(\phi_{N_0} \notin \mathcal{B}_{\rho/2} \mid \phi_0 \in \mathcal{B}_\rho) \le \frac{4}{\rho^2}\left(V_0 e^{-\mu_{\lambda_c}\alpha_0 N_0} + C\alpha_0(\beta_0^2 + \alpha_0\sigma^2)\right),$$

*where $V_0 = \|\phi_0 - \phi^*\|^2$. Hence for any target failure probability $\delta$, choosing $N_0 = O((\mu_{\lambda_c}\alpha_0)^{-1}\log(V_0/(\delta\rho^2)))$ and $\alpha_0$ small enough gives phase-2 entry with probability at least $1-\delta$. Conditional on this entry event, phase 2 converges almost surely to $\phi^*$ by Theorem 4.2.*

**Corollary 4.5** (Basin-entry speedup)**.** *Under the drift-improvement condition of Theorem 4.3, the leading phase-1 horizon scales as $N_0(\lambda) = O(1/(\alpha_0\mu_\lambda))$. Thus, relative to PG, Meta-MAPG shaping reduces the certified basin-entry horizon by the factor*

$$\frac{N_0(0)}{N_0(\lambda)} \gtrsim 1 + \frac{\lambda\mu_M}{\mu}$$

*up to higher-order local-linearisation terms. The condition $\mu_M > 0$ is game-dependent; if it fails, opponent-aware shaping need not enlarge the basin.*

**Theorem 4.6** (Restart globalisation of two-phase Meta-MAPG)**.** *Let each restart epoch run the two-phase procedure of Theorem 4.4. Let $\mathcal{B}_\rho \subset \mathcal{K}$ be the phase-1 local region and let the restart distribution $\nu$ satisfy $p_\rho := \nu(\mathcal{B}_\rho) > 0$. If epochs are independent, then the process enters a successful phase-2 epoch after finitely many restarts almost surely. If $T$ is the index of the first successful epoch, then*

$$\Pr(T > k) \le (1 - p_\rho(1-\delta))^k, \qquad \mathbb{E}[T] \le \frac{1}{p_\rho(1-\delta)}.$$

*Thus, after finitely many restarts, the process enters an epoch where the local theorem applies and then converges almost surely to a stable Nash equilibrium in $\mathcal{N}$. This globalises a local theorem by search; it does not make the vector field globally contracting. In the worst case, if $\nu$ is roughly uniform on a radius-$R$ $d$-dimensional compact set and $\mathcal{B}_\rho$ contains a radius-$\rho$ ball, the bound scales as $O((R/\rho)^d)$. High-dimensional domains therefore need structured or warm-start restarts.*

## 5. Proof Sketch

**Step 1: estimator decomposition.** This is the Meta-MAPG-specific step. The theorem of Kim et al. (2021) splits the sampled gradient into current, own-learning, and peer-learning score-function terms. Bounded rewards and scores give finite conditional variance; subtracting the conditional mean gives martingale noise. Finite unrolls and approximate opponent models give the bias $b_n$, controlled by Assumption 3.

**Step 2: summability.** The bias and variance terms are checked against the stochastic-approximation conditions. Because $\sum_n \alpha_n^2 < \infty$, the square-integrable martingale noise is summable in the Robbins–Siegmund sense (Robbins & Monro, 1951; Robbins & Siegmund, 1971; Hall & Heyde, 1980). Because $\sum_n \alpha_n\beta_n < \infty$, the deterministic truncation bias is also summable.

**Step 3: local theorem transfer.** Map the Meta-MAPG update to Giannou et al.'s local PG template. In the annealed regime, the limiting field is Nash-consistent: local stationary points are the same stable Nash policies as for the base game-gradient field. The Lyapunov recursion is the same as in Giannou et al. (2022), up to summable forcing terms.

**Step 4: two-phase transfer.** Theorem 4.4 separates finite-time shaping from asymptotic convergence. Phase 1 uses constant shaping only for basin entry, with drift constant $\mu_{\lambda_c}$ from Theorem 4.3. After entry into $\mathcal{B}_{\rho/2}$, the schedule anneals; Theorem 4.2 follows from the local attraction theorem for stable SOS Nash policies. No global monotonicity is required. The argument is local, as in Giannou et al.

**Step 5: restart globalisation.** Restarts globalise the local result by a geometric argument. Each epoch is independent. The restart distribution assigns mass $p_\rho$ to the phase-1 region, and the two-phase theorem reaches phase 2 with probability at least $1-\delta$. Each epoch therefore succeeds with probability at least $p_\rho(1-\delta)$. The first successful epoch is geometrically dominated. Basin maps in Section 6 show how opponent-aware shaping can improve this constant by increasing basin-entry probability.

## 6. Experiments

The experiments are small by design. We use tabular Bernoulli policies in Stag Hunt and finite-horizon iterated Prisoner's Dilemma (IPD). Stag Hunt is the core test: two stable Nash equilibria, non-trivial basins, and a two-dimensional policy space where the drift constants $\mu$ and $\mu_M$ in Theorem 4.3 can be measured. It is also the standard testbed for LOLA, SOS, and COLA (Foerster et al., 2018a; Letcher et al., 2019; Willi et al., 2022). IPD tests the same estimator in a finite-horizon sequential game. Our Stag Hunt is a one-state, one-step, two-agent Markov game. Each agent samples $C$ or $D$ from a Bernoulli policy, with row-player payoff matrix

| | $C$ | $D$ |
|---|---|---|
| $C$ | 4 | 0 |
| $D$ | 3 | 2 |

and the column-player payoff given by the transpose. Thus $(C, C)$ is payoff-dominant with social welfare 8, while $(D, D)$ is risk-dominant with social welfare 4. In the figures, each coordinate is an agent's cooperation probability $p_i = \pi_i(C)$. Success means both final cooperation probabilities exceed 0.82.

Updates are sample-based: for a trajectory score $s_i = \nabla_{\phi_i}\log p_\phi(\tau)$ and return $R_i$, the base estimator is $\hat{g}_i =$

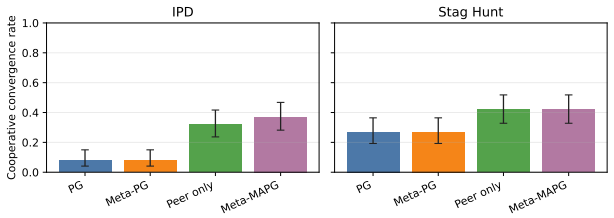


*Figure 1.* Component ablation with Wilson 95% confidence intervals over 100 seeds. Peer-aware methods (peer-only, Meta-MAPG) separate from non-peer methods (PG, Meta-PG) in both games. Peer-only and full Meta-MAPG are not distinguishable within CIs; the claim is the peer-learning effect, not strict dominance of the full method.

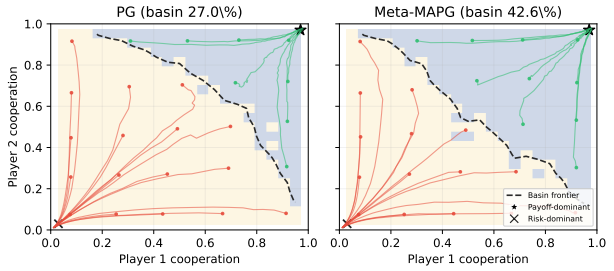


*Figure 2.* Basin geometry and dynamics in Stag Hunt. Background shading is the empirical basin map on a $21 \times 21$ grid (lavender = payoff-dominant, cream = risk-dominant). The dashed contour is the empirical separatrix. Overlaid trajectories start from a $5 \times 5$ sub-grid; green curves succeed, red curves fail. Meta-MAPG pushes the frontier left and down: the cooperative basin grows from 27.0% under PG to 42.6%, consistent with Theorem 4.3.

$\frac{1}{B}\sum_b R_i^{(b)} s_i^{(b)}$. The own-learning term uses the likelihood-ratio Hessian estimator

$$\widehat{H}_{ii} = \frac{1}{B}\sum_b R_i^{(b)} \left( s_i^{(b)} s_i^{(b)\top} + \nabla^2_{\phi_i} \log p_\phi(\tau^{(b)}) \right),$$

and the peer-learning term uses the cross-score estimator

$$\widehat{C}_{ji} = \frac{1}{B}\sum_b R_j^{(b)} s_j^{(b)} s_i^{(b)\top}.$$

The implemented one-step correction is $\eta \widehat{H}_{ii}^\top \hat{g}_i$ for own learning and $\eta \widehat{C}_{ji}^\top \widehat{\nabla_{\phi_j} J_i}$ for peer learning. We compare PG, Meta-PG (no peer term), peer-only LOLA-style shaping (no own term), and full Meta-MAPG. Unless stated otherwise, summaries use 100 seeds; restart summaries pair initial policies across all four methods; restart-budget curves pair PG and Meta-MAPG starts; all main experiments use the same peer coefficient. Raw CSVs and run configuration are included.

The ablation shows that peer-learning terms matter; it does not show that full Meta-MAPG dominates every LOLA-style variant. The trajectory figure shows the finite-time

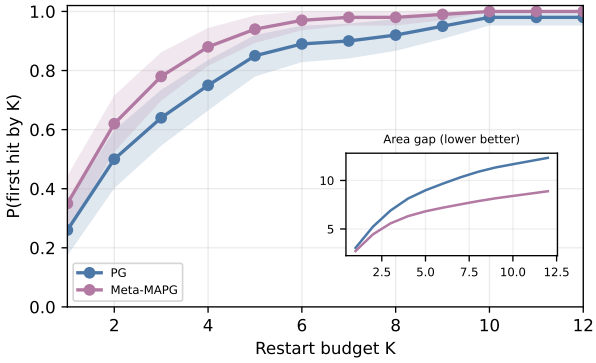


*Figure 3.* Restart budget as equilibrium-selection budget in Stag Hunt. With paired restart initialisations, we plot the probability of discovering the payoff-dominant equilibrium by budget $K$; bands show 95% confidence intervals. The inset shows $\sum_{k \le K}(8 - W_k)$, the cumulative gap below best welfare. Lower is better. Meta-MAPG reaches the good basin with fewer restarts.

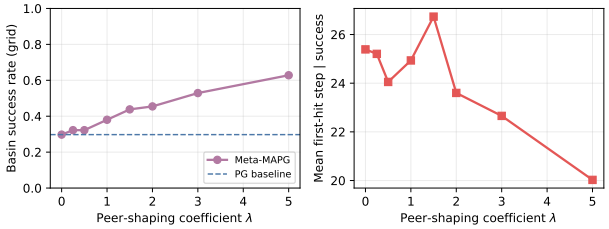


*Figure 4.* Peer-shaping sweep on Stag Hunt. Left: payoff-dominant basin fraction on an $11 \times 11$ initial-policy grid as peer coefficient $\lambda$ varies. The rate grows from the PG baseline (29.8%) to 62.8% at $\lambda = 5$, roughly linearly for small $\lambda$, as predicted by $\rho_\lambda \gtrsim \rho_0 + \lambda\mu_M/(2L)$. Right: conditional mean first-hit step contracts from $\approx 25$ to $\approx 20$, matching Theorem 4.5.

mechanism: opponent-aware shaping changes the local flow where the basin map changes. The restart-budget, basin, and peer-sweep experiments support Theorems 4.3 to 4.6: restarts turn local convergence into equilibrium search, and shaping improves that search by increasing cooperative basin mass and first-passage probability. Figure 5 runs the two-phase schedule directly. It shows that annealing after $N_0$ steps preserves basin-entry performance while restoring the Nash target. The MLP-policy IPD check in Section G is a transfer check, not a deep-MARL benchmark. The estimator sanity check in Section H is not a proof of Assumption 2; it checks that the sample estimator moves in the predicted direction as batch size grows.

## 7. Discussion

The theorem is local first. Meta-MAPG inherits Giannou et al.'s stable-Nash theory only inside attracting neighbourhoods and only after the estimator has stochastic-approximation form. The two-phase theorem fixes the algo-

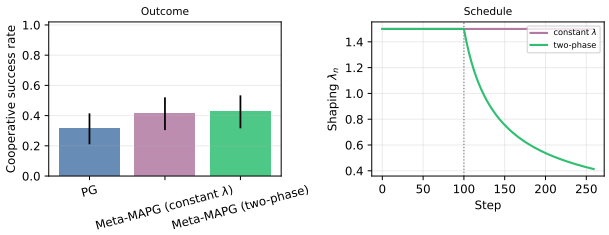


*Figure 5.* Two-phase schedule vs constant shaping in Stag Hunt (80 seeds, 260 steps, $\lambda_c = 1.5$). PG reaches 31%, constant-shaping Meta-MAPG reaches 41%, and the two-phase schedule reaches 43%. Annealing restores Nash consistency without losing basin-entry performance. Right: $\lambda_n$ over time; the dotted line marks $N_0 = 100$.

*Table 2.* Main 100-seed summary. Cooperative success uses a minimum initial-state cooperation threshold of 0.82. Restart success and mean restarts are reported for all methods using paired restart initialisations.

| Game | Method | Ablation success | Restart success | Restarts |
|---|---|---|---|---|
| stag hunt | PG | 27% | 100% | 1.71 |
| stag hunt | Meta-PG | 27% | 100% | 1.75 |
| stag hunt | Peer only | 42% | 100% | 1.24 |
| stag hunt | Meta-MAPG | 42% | 100% | 1.24 |
| ipd | PG | 8% | 51% | 8.68 |
| ipd | Meta-PG | 8% | 57% | 7.90 |
| ipd | Peer only | 32% | 97% | 2.91 |
| ipd | Meta-MAPG | 37% | 100% | 2.13 |

rithmic role: constant shaping is for basin entry; annealing restores the original Nash target. Restarts search across basins. Selecting the best endpoint makes restart budget an equilibrium-selection resource. Its cost depends on basin probability $p_\rho$; shaping helps only when it increases that probability.

The assumptions are real. The estimator must have bounded conditional variance. Truncation must make unroll bias summable. Asymptotic shaping must be Nash-consistent. Constant shaping is useful for basin geometry, but a constant peer term can move fixed points in general-sum games. This is why the theorem separates annealed convergence from constant-shaping diagnostics.

The experiments are deliberately limited. Matrix and tabular games let us measure the theorem's objects: basins, restarts, drift, and estimator variance. Naive random restarts degrade with dimension: a radius-$\rho$ basin inside a radius-$R$ search region has worst-case mass $(\rho/R)^d$. Deep MARL needs stronger variance control, opponent models, differentiable longer unrolls, and structured restarts such as warm starts around high-welfare policies, possibly with DiCE-style estimators (Foerster et al., 2018b). The claim here is narrower: opponent-aware gradients can fit stochastic approximation, and basin-entry efficiency is the right global metric when convergence is local.

We do not claim unconditional convergence of arbitrary opponent-aware learning in deep MARL. We show that finite-unroll Meta-MAPG fits classical stochastic approximation under boundedness, smoothness, Nash-consistency, and summability assumptions. Independent restarts globalise the local result when local basin mass is positive. The MLP-policy check in Section G only tests whether the peer-vs-nonpeer separation survives a modest neural parameterisation.

## A. Detailed Estimator Decomposition

For a sampled trajectory $\tau$, write $S_i(\tau) = \nabla_{\phi_i} \log p_\phi(\tau)$ and $H_i(\tau) = \nabla^2_{\phi_i} \log p_\phi(\tau)$. The base likelihood-ratio estimator is $R_i(\tau)S_i(\tau)$ (Williams, 1992; Sutton & Barto, 2018). The one-step own-learning derivative follows from differentiating $J_i(\phi_i + \eta\nabla_{\phi_i} J_i, \phi_{-i})$:

$$\nabla_{\phi_i} J_i + \eta \left[\nabla^2_{\phi_i} J_i\right]^\top \nabla_{\phi_i} J_i.$$

The Hessian identity gives

$$\nabla^2_{\phi_i} J_i = \mathbb{E}\left[R_i(\tau)\left(S_i(\tau)S_i(\tau)^\top + H_i(\tau)\right)\right],$$

which is the estimator used in the experiments.

For the peer term, let agent $j$ take one inner step $\phi'_j = \phi_j + \eta\nabla_{\phi_j} J_j$. Differentiating $J_i(\phi_i, \phi'_j)$ with respect to $\phi_i$ contributes

$$\eta \left[\nabla_{\phi_i}\nabla_{\phi_j} J_j\right]^\top \nabla_{\phi_j} J_i.$$

The mixed likelihood-ratio identity gives

$$\nabla_{\phi_i}\nabla_{\phi_j} J_j = \mathbb{E}\left[R_j(\tau)S_j(\tau)S_i(\tau)^\top\right],$$

which is the sample cross-score matrix used for the peer correction. Longer Meta-MAPG unrolls repeat this chain-rule structure across $L$ inner updates; bounded derivatives on $\mathcal{K}$ allow the omitted tail and finite-modeling error to be collected in $b_n$.

## B. Local Lyapunov Drift

We spell out the stochastic-approximation step behind Theorem 4.2. The goal is not to reprove Giannou et al. (2022); it is to show that Meta-MAPG has the same Lyapunov form after decomposition.

**Lemma B.1** (One-step local Lyapunov drift)**.** *Fix an SOS Nash equilibrium $\phi^*$ and a ball $\mathcal{B}_\rho = \{\phi : \|\phi - \phi^*\| \le \rho\}$ on which the effective field $F$ satisfies*

$$\langle F(\phi), \phi - \phi^* \rangle \le -\mu\|\phi - \phi^*\|^2.$$

*Suppose the stopped iterates remain in $\mathcal{B}_\rho$ and the Meta-MAPG update admits*

$$g_n = F(\phi_n) + \xi_{n+1} + b_n,$$
$$\mathbb{E}[\xi_{n+1} \mid \mathcal{F}_n] = 0, \qquad \mathbb{E}[\|\xi_{n+1}\|^2 \mid \mathcal{F}_n] \le \sigma^2,$$

*with* $\|b_n\| \le \beta_n$. *If* $V_n = \|\phi_n - \phi^*\|^2$ *and* $\|F(\phi)\| \le C_F$ *on* $\mathcal{B}_\rho$, *then*

$$\mathbb{E}[V_{n+1} \mid \mathcal{F}_n] \le (1 - 2\mu\alpha_n)V_n + 2\alpha_n\beta_n\sqrt{V_n} + 3\alpha_n^2(C_F^2 + \beta_n^2 + \sigma^2). \tag{3}$$

*Proof.* Projection onto the convex compact set $\mathcal{K}$ is non-expansive, so it is enough to bound the unprojected step. Expanding the square gives

$$\begin{aligned} V_{n+1} &\le \|\phi_n - \phi^* + \alpha_n g_n\|^2 \\ &= V_n + 2\alpha_n\langle g_n, \phi_n - \phi^*\rangle + \alpha_n^2\|g_n\|^2. \end{aligned}$$

Taking conditional expectation and using $\mathbb{E}[\xi_{n+1} \mid \mathcal{F}_n] = 0$,

$$\begin{aligned} \mathbb{E}[\langle g_n, \phi_n - \phi^*\rangle \mid \mathcal{F}_n] &= \langle F(\phi_n), \phi_n - \phi^*\rangle \\ &\quad + \langle b_n, \phi_n - \phi^*\rangle \\ &\le -\mu V_n + \beta_n\sqrt{V_n}. \end{aligned}$$

For the quadratic term, $(a+b+c)^2 \le 3(a^2+b^2+c^2)$ gives

$$\begin{aligned} \mathbb{E}[\|g_n\|^2 \mid \mathcal{F}_n] &\le 3\big(\|F(\phi_n)\|^2 + \|b_n\|^2 \\ &\quad + \mathbb{E}[\|\xi_{n+1}\|^2 \mid \mathcal{F}_n]\big) \\ &\le 3(C_F^2 + \beta_n^2 + \sigma^2). \end{aligned}$$

Substituting the last two displays into the square expansion proves (3). □

If $\sum_n \alpha_n\beta_n < \infty$ and $\sum_n \alpha_n^2 < \infty$, then (3) is in Robbins–Siegmund form. Indeed,

$$2\alpha_n\beta_n\sqrt{V_n} \le \alpha_n\beta_n(1 + V_n),$$

so for all large $n$,

$$\mathbb{E}[V_{n+1} \mid \mathcal{F}_n] \le (1 - a_n)V_n + \eta_n$$

with $\sum_n a_n = \infty$ and $\sum_n \eta_n < \infty$. Robbins–Siegmund yields $V_n \to V_\infty$ almost surely and $\sum_n a_n V_n < \infty$; since $\sum_n a_n = \infty$, $V_\infty = 0$. Thus $\phi_n \to \phi^*$ almost surely on the local event.

## C. Proof of Proposition 4.3

We prove the two parts separately.

**Linearised drift bound.** By assumption $v, M \in C^1$ in a neighbourhood of $\phi^*$. Both $J_v$ and $J_M$ are real matrices; their symmetric parts $\mathrm{sym}(A) := (A + A^\top)/2$ are Hermitian. By definition, $-\lambda_{\max}(\mathrm{sym}(J_v)) = \mu$ (the SOS margin) and $-\lambda_{\max}(\mathrm{sym}(J_M)) = \mu_M$. For any $\lambda \ge 0$,

$$\mathrm{sym}(J_\lambda) = \mathrm{sym}(J_v) + \lambda\,\mathrm{sym}(J_M),$$

which is Hermitian. Weyl's inequality for Hermitian matrices gives

$$\lambda_{\max}\big(\mathrm{sym}(J_\lambda)\big) \le \lambda_{\max}\big(\mathrm{sym}(J_v)\big) + \lambda\,\lambda_{\max}\big(\mathrm{sym}(J_M)\big).$$

Negating both sides yields (2). The bound is exact, not asymptotic, and requires no smallness condition on $\lambda$.

**Certified radius.** Let $\mu_\lambda := \mu + \lambda\mu_M$. Fix $\phi \in \mathcal{B}_\rho(\phi^*)$ and write $x := \phi - \phi^*$. Because $F_\lambda(\phi^*) = v(\phi^*) + \lambda M(\phi^*) = 0$ and $M(\phi^*) = 0$, Taylor's theorem with the integral remainder gives

$$F_\lambda(\phi) = J_\lambda x + R_\lambda(x), \qquad \|R_\lambda(x)\| \le \tfrac{1}{2}L_\lambda\|x\|^2,$$

where $L_\lambda$ is the Lipschitz constant of $DF_\lambda$ on $\mathcal{B}_\rho(\phi^*)$. Taking inner products,

$$\langle F_\lambda(\phi), x\rangle = \langle J_\lambda x, x\rangle + \langle R_\lambda(x), x\rangle.$$

The first term obeys $\langle J_\lambda x, x\rangle = x^\top \mathrm{sym}(J_\lambda)x \le -\mu_\lambda\|x\|^2$ by the Rayleigh quotient and (2). The remainder obeys $|\langle R_\lambda(x), x\rangle| \le \frac{1}{2}L_\lambda\|x\|^3$. Therefore

$$\langle F_\lambda(\phi), \phi - \phi^*\rangle \le -\mu_\lambda\|x\|^2 + \tfrac{1}{2}L_\lambda\|x\|^3.$$

Whenever $\|x\| \le \rho_\lambda := \mu_\lambda/(2L_\lambda)$, the cubic term is at most $(\mu_\lambda/4)\|x\|^2$, so

$$\langle F_\lambda(\phi), \phi - \phi^*\rangle \le -\tfrac{3}{4}\mu_\lambda\|x\|^2 \le -\tfrac{1}{2}\mu_\lambda\|x\|^2.$$

This proves the stated SOS drift on $\mathcal{B}_{\rho_\lambda}(\phi^*)$. □

## D. Proof of Theorem 4.4

Write $V_n = \|\phi_n - \phi^*\|^2$ and let $\tau := \inf\{n \le N_0 : \phi_n \notin \mathcal{B}_\rho\}$ with $\tau = N_0 + 1$ if the iterates never leave. On the event $\{n < \tau\}$ the phase-1 drift of Theorem 4.3 applies with constant $\mu_{\lambda_c}$.

**Step 1: one-step drift.** On $\{n < \tau\}$, by Assumption 2 and the phase-1 drift bound,

$$\begin{aligned} \mathbb{E}[V_{n+1} \mid \mathcal{F}_n] &= \mathbb{E}\big[\|\phi_n - \phi^* + \alpha_0 g_n\|^2 \mid \mathcal{F}_n\big] \\ &= V_n + 2\alpha_0\langle F_{\lambda_c}(\phi_n), \phi_n - \phi^*\rangle \\ &\quad + 2\alpha_0\langle b_n, \phi_n - \phi^*\rangle + \alpha_0^2\mathbb{E}[\|g_n\|^2 \mid \mathcal{F}_n] \\ &\le V_n - 2\alpha_0\mu_{\lambda_c}V_n + 2\alpha_0\beta_0\sqrt{V_n} + \alpha_0^2 G^2, \end{aligned}$$

where $G^2 := 3(C_{F_{\lambda_c}}^2 + \beta_0^2 + \sigma^2)$ is a uniform bound on the conditional second moment of $g_n$ (with $C_{F_{\lambda_c}}$ the sup-norm of $F_{\lambda_c}$ on $\mathcal{B}_\rho$). Applying $2ab \le \eta a^2 + \eta^{-1}b^2$ with $a = \sqrt{V_n}$, $b = \beta_0$, and $\eta = \mu_{\lambda_c}$,

$$\mathbb{E}[V_{n+1} \mid \mathcal{F}_n] \le (1 - \mu_{\lambda_c}\alpha_0)V_n + \frac{\alpha_0\beta_0^2}{\mu_{\lambda_c}} + \alpha_0^2 G^2.$$

**Step 2: unconditional recursion on the stopped process.** Define the stopped sequence $\tilde{V}_n := V_{n\wedge\tau}$. The drift bound and the fact that $\tilde{V}_{n+1} = \tilde{V}_n$ on $\{n \geq \tau\}$ give

$$\mathbb{E}[\tilde{V}_{n+1}] \leq (1 - \mu_{\lambda_c}\alpha_0)\mathbb{E}[\tilde{V}_n] + c_1,$$
$$c_1 := \frac{\alpha_0\beta_0^2}{\mu_{\lambda_c}} + \alpha_0^2 G^2.$$

Iterating,

$$\mathbb{E}[\tilde{V}_{N_0}] \leq (1 - \mu_{\lambda_c}\alpha_0)^{N_0} V_0 + \frac{c_1}{\mu_{\lambda_c}\alpha_0}.$$

Using $(1-x)^k \leq e^{-xk}$ for $x \in [0,1]$,

$$\mathbb{E}[\tilde{V}_{N_0}] \leq V_0\, e^{-\mu_{\lambda_c}\alpha_0 N_0} + \frac{\beta_0^2}{\mu_{\lambda_c}^2} + \frac{\alpha_0 G^2}{\mu_{\lambda_c}}.$$

**Step 3: Markov's inequality.** If $\phi_{N_0} \notin \mathcal{B}_{\rho/2}$ and $\tau > N_0$, then $V_{N_0} \geq (\rho/2)^2$. If $\tau \leq N_0$, by continuity of trajectories (or by projection onto $\mathcal{K}$) the iterate at time $\tau$ already lies outside $\mathcal{B}_\rho$, and in particular $V_\tau \geq \rho^2 \geq (\rho/2)^2$; $\tilde{V}_{N_0} = V_\tau$ in that case. Hence in both cases $\tilde{V}_{N_0} \geq (\rho/2)^2$. Therefore

$$\Pr(\phi_{N_0} \notin \mathcal{B}_{\rho/2} \mid \phi_0 \in \mathcal{B}_\rho)$$
$$\leq \Pr(\tilde{V}_{N_0} \geq (\rho/2)^2) \leq \frac{4\mathbb{E}[\tilde{V}_{N_0}]}{\rho^2}.$$

Combining with Step 2,

$$\Pr(\phi_{N_0} \notin \mathcal{B}_{\rho/2} \mid \phi_0 \in \mathcal{B}_\rho)$$
$$\leq \frac{4}{\rho^2}\left(V_0 e^{-\mu_{\lambda_c}\alpha_0 N_0} + C\alpha_0(\beta_0^2 + \alpha_0\sigma^2)\right),$$

for a constant $C$ depending on $\mu_{\lambda_c}, C_{F_{\lambda_c}}$.

**Step 4: choice of $N_0$.** Given a target $\delta \in (0,1)$, set

$$N_0 = \left\lceil \frac{1}{\mu_{\lambda_c}\alpha_0} \log\frac{8V_0}{\delta\rho^2} \right\rceil, \qquad \alpha_0 \leq \frac{\delta\rho^2}{8C(\beta_0^2 + \sigma^2)}.$$

Each summand in Step 3 is then at most $\delta/2$, so the total failure probability is at most $\delta$.

**Step 5: phase 2.** Conditional on $\phi_{N_0} \in \mathcal{B}_{\rho/2}$, the subsequent recursion uses $\alpha_n = \alpha_0 n^{-p}$ and $\lambda_n = \lambda_c n^{-q}$ with $1/2 < p \leq 1$ and $q > 1 - p$, so $\sum_n \alpha_n = \infty$, $\sum_n \alpha_n^2 < \infty$, and $\sum_n \alpha_n\lambda_n < \infty$ by direct integral comparison. Together with Assumptions 2, 3 and 5, this satisfies the hypotheses of Theorem 4.2 on the event $\{\phi_{N_0} \in \mathcal{B}_{\rho/2}\}$. Conditional on this event, $\phi_n \to \phi^*$ almost surely. The total probability that the two-phase procedure converges in a single epoch is therefore at least $(1-\delta)\cdot 1 = 1 - \delta$. □

## E. Proof of Theorem 4.6

Let $\mathcal{F}_k^{\text{epoch}}$ denote the $\sigma$-algebra generated by epochs $1, \ldots, k-1$ (outcomes only; restart draws are independent by assumption). Let $A_k$ be the event that epoch $k$ starts in $\mathcal{B}_\rho$ and that the two-phase procedure of Theorem 4.4 reaches phase 2 and converges within that epoch.

**Step 1: per-epoch lower bound.** Restart $k$ samples $\phi_0^{(k)} \sim \nu$ independently of $\mathcal{F}_k^{\text{epoch}}$, so

$$\Pr(\phi_0^{(k)} \in \mathcal{B}_\rho \mid \mathcal{F}_k^{\text{epoch}}) = \nu(\mathcal{B}_\rho) = p_\rho.$$

Conditional on $\phi_0^{(k)} \in \mathcal{B}_\rho$, Theorem 4.4 gives $\Pr(\text{epoch succeeds}) \geq 1 - \delta$. Therefore

$$\Pr(A_k \mid \mathcal{F}_k^{\text{epoch}}) \geq p_\rho(1-\delta) =: p_\star > 0.$$

**Step 2: geometric tail.** Let $T := \inf\{k \geq 1 : A_k \text{ occurs}\}$. Then

$$\Pr(T > k) = \mathbb{E}\left[\prod_{j=1}^{k} \mathbf{1}_{A_j^c}\right]$$
$$= \mathbb{E}\left[\mathbb{E}[\mathbf{1}_{A_k^c} \mid \mathcal{F}_k^{\text{epoch}}] \prod_{j=1}^{k-1} \mathbf{1}_{A_j^c}\right].$$

By Step 1, $\mathbb{E}[\mathbf{1}_{A_k^c} \mid \mathcal{F}_k^{\text{epoch}}] \leq 1 - p_\star$ almost surely, so iterating,

$$\Pr(T > k) \leq (1 - p_\star)^k.$$

Hence $\Pr(T < \infty) = 1$ and

$$\mathbb{E}[T] = \sum_{k\geq 0} \Pr(T > k) \leq \sum_{k \geq 0} (1 - p_\star)^k = \frac{1}{p_\star}.$$

**Step 3: dimensional dependence.** If $\nu$ is lower-bounded by density $m > 0$ on a region containing $\mathcal{B}_\rho(\phi^*)$ in $\mathbb{R}^d$, then $p_\rho \geq m\, c_d \rho^d$ where $c_d = \pi^{d/2}/\Gamma(d/2+1)$ is the volume of the unit $d$-ball. Combined with the certified radius $\rho_\lambda \geq (\mu + \lambda\mu_M)/(2L_\lambda)$ from Theorem 4.3, the expected number of epochs obeys

$$\mathbb{E}[T] \leq \frac{1}{(1-\delta)mc_d}\left(\frac{2L_\lambda}{\mu + \lambda\mu_M}\right)^d.$$

Shaping reduces the base of this dimensional factor whenever $\mu_M > 0$; the overall worst-case dependence is still exponential in $d$, as discussed in Section 4. □

## F. Experiment Reproducibility

The experiment scripts, figures, CSVs, and run configurations are included. The tabular script saves its command-line configuration and outputs under `artifacts/main`.

These files cover the ablation, restart, trajectory, basin, peer-sweep, annealing, and estimator-sanity figures. Updates use no exact gradients; expected returns are computed only for logging and evaluation.

The optional MLP-policy IPD check in Section G requires PyTorch and uses DiCE-style surrogate losses for the higher-order terms. Its aggregate results are stored under `artifacts/mlp`. The repository README contains the full commands for regenerating both sets of experiments.

## G. Non-Tabular IPD Check

To check that the ablation is not purely tabular, we repeat IPD with a two-layer MLP policy for each agent: a 5-dimensional one-hot previous-action input, two tanh hidden layers of width 16, and one Bernoulli cooperation logit. This scales the policy from 5 tabular logits to about 385 parameters per agent. We keep the tabular budget, batch size, seed schedule, peer coefficient, and own coefficient. Higher-order terms use PyTorch autograd with DiCE-style surrogate losses (Foerster et al., 2018b).

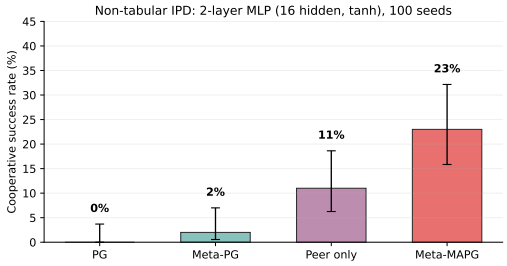


*Figure 6.* Non-tabular IPD check with MLP policies. Cooperative success rate over 100 seeds with Wilson 95% confidence intervals. Peer-aware methods remain above non-peer methods under the same coefficients as the tabular experiments: PG reaches 0%, Meta-PG 2%, peer-only 11%, and Meta-MAPG 23%. This is a transfer check, not a deep-MARL benchmark.

## H. Estimator Sanity Check

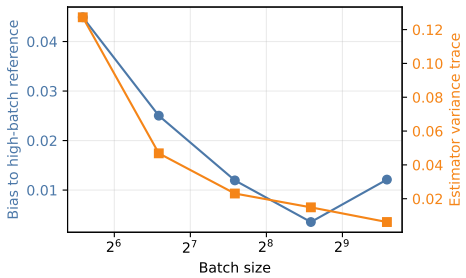


*Figure 7.* Estimator sanity check. At a fixed Stag Hunt parameter, the variance trace of the Meta-MAPG estimator decreases from 0.1272 at batch size 48 to 0.0062 at batch size 768. Bias is measured relative to a high-batch reference estimator, not an exact analytic gradient.

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